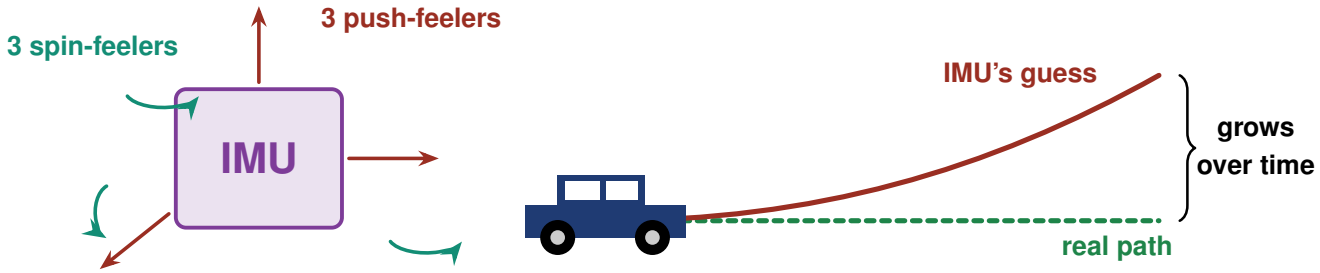
The IMU: Feeling Motion

BIG IDEA

An IMU feels every push and turn, but its guess slowly drifts.



THE IDEA

When a car speeds up, you get pushed back into your seat. When it turns, you lean to the side. You feel the motion even with your eyes closed. An IMU is a little box that feels motion the same way. By adding up all those feelings many times a second, it can guess where the car went, even inside a tunnel. But like walking with your eyes closed, the guess slowly gets worse.

REMEMBER

- ✓ An IMU has three push-feelers and three spin-feelers.
- ✓ It adds up thousands of tiny readings, so tiny mistakes add up too.
- ✓ That is why the IMU needs help from GPS or the laser to stay on track.

TRY THIS

In an open, safe space, with a friend watching, close your eyes and walk 10 steps toward a spot, turn around and walk back. How far from the start did you end up?

The full module is in the Reference Book.